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Logical operators

- The following operators are called logical operators.

Not (Negation)
And (Conjunction)
Or (Disjuaction)
if/if...then (implication)

- The Not is a unary operator while all the others are binary operators.

Compound positions

- compound proposition are formed by using only logical connectives and primitive propositions. Typically, compound propositions do not contain variables or quantifiers.

- Compound propositions are also propositions (How??)

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propositions

- A proposition (also called as statement) is a sente that can be true or false. However, not both at the same time simultaneously.

eg:- "Saman is awake" is a proposition. because at any given time either Saman is not awake or Saman is awake, and saman can not be both awake and asleep at the same time.

2. "Wake up!" is not a proposition because it cannot true or false.

Atomic propositions

A statement that cannot be broken down into small statements, is known as the atomic propositions (or primitive propositions)

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Rules for connectives

The truth value of a compound statement is determined by the truth values of its atomic parts together with applying the following rules for the connectives.

- $p \wedge q$ is called the conjunction of p and q. It is pronounced as p and q. $p \wedge q$ is true when both p and q are true. and false otherwise.

- $p \vee q$ called the disjunction of p and q. It is pronounced p or q.

- $p \rightarrow q$ is called implication. It is pronounced if p then q or p implies q. $p \rightarrow q$ is true when p is false or q is true (or both). And it is false when p is true and q is false.

- $\neg p$ is called the negation of p. It is pronounced not p.

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Truth table for negation

p	¬p
T	F
F	T

- Negation is so called UNARY opera as it only applies to one propositio

- Other boolean operations are binary. In binary Boolean operations there are two atomic propositions.

Truth table for conjunction

p	q	p∧q
T	T	T
T	F	F
F	T	F
F	F	F

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Truth table for disjunction

p	q	p∨q
T	T	T
T	F	T
F	T	T
F	F	F

Truth table for implication (→)

- In binary boolen operations there are two atomic propositions. Each of the propositions can be either true or false.

p	q	p→q
T	T	T
T	F	F
F	T	T
F	F	T

- The implication of a boolean expression p with a second boolean expression q is false only when p is true.

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precedence of logical operators

	precedence	operator symbol
-Not [Negation]	1	$\neg$, $\sim$
. And [Conjunction]	2	$\wedge$, &
Or [Disjunction]	3	V
if/if...then [Implication]	4	$\rightarrow$, $\Rightarrow$

Computing Truth Tables

- We consider formulae computationally. i.e given a formula C, we want to compute its truth table.

- Let a formula C consist of the atomic propositions $p_1, p_2, \ldots, p_n$.

- To compute the truth table do the following.

1) List its atomic propositions $p_1, p_2 \ldots p_n$ and count their number, n.

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2) Form a table with $m = 2^n$ rows.

3) List all possible combinations of the truth values for $p_1, p_2, \ldots, p_n$.

4) Prioritize the logical operations in C, finding main logic operation.

5) Compute these operations in order.

1) $(p \to p) \to p$

p	$p \to p$	$(p \to p) \to p$
T	T	T
F	T	T

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Set - Definition

- A set is a well defined collection of distinct objects or elements.

- The items of a set are refferred to as members or elements of the set.

- The number of elements in a set can be either finite or infinite.

- When the number of elements in a set is finite such a set is called finite set and when the number of elements of a set is infinite the set is called an ifinite set.

- $a \in A$ - is a is an element of the set A. Represent as $|A|$

- example: Let $A = \{a, b, c\}$, Then the cardinality of set $A = |A| = 3$

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Definition

- Any well-defined collection of distinct objects is called a set.

① A = {'Amal', 'Bimal', 'Carol', 'Dias', 'Emma'} is a set of names.

② B = {1, 3, 5, 7, 9} is a set of odd natural numbers between 1-10.

③ C = {2, 4, 6, 8, 10} is the set of even natural numbers between 1-10.

- Null set (Empty set) → Is a set with no elements
  Notation : ϕ , {}
  (Note {0} is not an empty set)

- **Equality of sets:**
  The sets A and B are equal if and only if A and B both contain the same elements.

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- **Subsets:**

A is a subset of B if every element of A is also in B. [A is a subset of itself]

Notation: $A \subseteq B$

- If A is a subset of B and $A \neq B$, then A is called a proper subset of B. → proper subset ($\subset$)

eg: $D = \{$ 'Amal', 'Bimal', 'Emma' $\}$. Then D is a subset of A

- **Universal set**

Denoted by $\mathcal{U}$, universal set is the set of all elements in a given in a discussion.

eg:- If A of eg 1 contains names from a given class, then $\mathcal{U}$ is the set of all names in the class.

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Intersection of sets:

- If A and B are two subsets of U, the 'intersection of A and B' is the set consisting of elements that belong to both A and B.

Notation: $A \cap B$

$B = \{1, 3, 5, 7, 9\}$, $E = \{1, 2, 3, 4, 5\}$

$B \cap E = \{1, 3, 5\}$

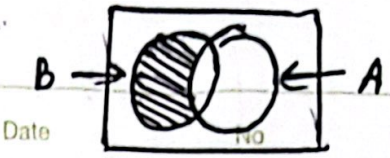
Complement of a set:

- If A is a subset of U, the set of elements of U not in A, is called the 'Complement of A'.
- Notation: A', $\bar{A}$, A^c

$B = \{1, 3, 5, 7, 9\}$ $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$B^A_c = \{2, 4, 6, 8, 10\}$

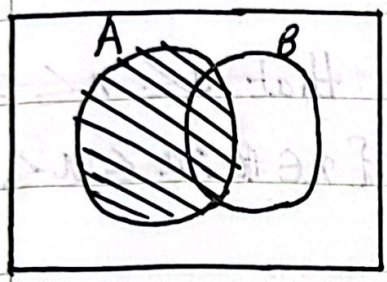
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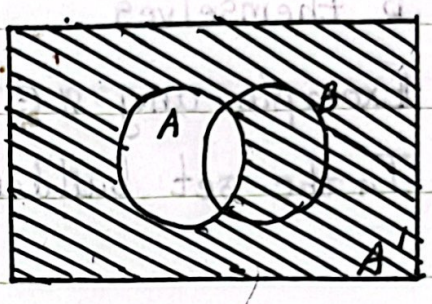
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Difference between B and A

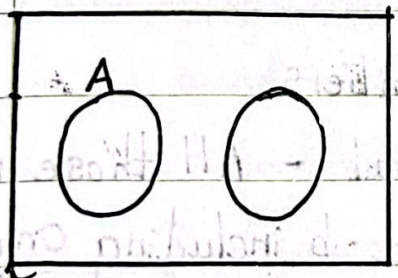
- If A and B are two subsets of U, 'B-A' is the set that consists of elements of B that is not A. (i.e $B-A = B \cap A^c$; Alternative notation : B/A)



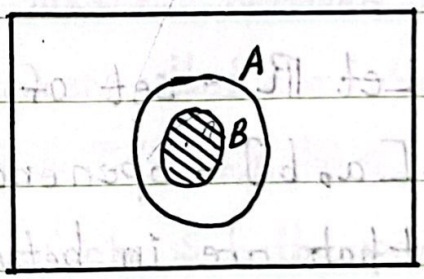
set A



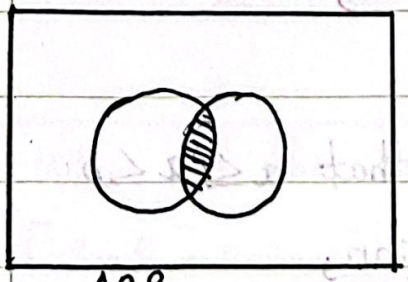
A is the complement of A



A and B are disjoint sets

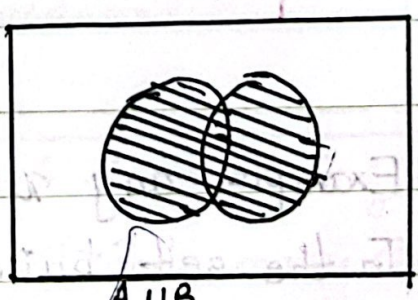


B is the proper subset of A



$A \cap B$

A and B are overlapping sets.



$A \cup B$

A and B are overlapping sets.

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Notation.

- Let $\mathbb{R}$ - set of real numbers.
- $[a, b]$ - closed interval - All those numbers that are in between a and b, including a and b themselves.
- Example any $x \in \mathbb{R}$ x such that $a \leq x \leq b$
- In the set builder notation $\{x \in \mathbb{R} : a \leq x \leq b\}$

Open ended interval $[a, b)$

- Let $\mathbb{R}$ - set of Real numbers.
- $[a, b)$ - openended interval - All those numb that are in between a and b including only a but not b



- Example any $x \in \mathbb{R}$ such that $a \leq x < b$
- In the set builder notation,
  $\{x \in \mathbb{R} : a \leq x < b\}$

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open beginning interval $(a, b]$

- Example any $x \in \mathbb{R}$ such that $a < x \leq b$

In the set builder notation $\{x \in \mathbb{R} : a < x \leq b\}$



Intervals - Summery

Let $x \in \mathbb{R}$ where $\mathbb{R}$ is a set of real numbers.

- $[a, b]$ - abbreviate a closed interval between numbers a and b. All those numbers that are between a and including a and b themselves.
- (a, b) - denotes an interval which is both open beginning and open ended. This set does not contain both and and b.

What are the cardinality of the following sets?

$[3, 4] = \infty$ (This set has infinitely many element

$\{3, 4\} = 2$

$[3, 4) = \infty$

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Whole Numbers = {0, 1, 2, 3, 4 ... }

- You have all the non-negative integers starting from 0 and continuing infinitely. Countable.

Counting Numbers (Natural numbers) = {1, 2, 3, 4, 5, ..}

- positive integers starting from 1 and continuing infinitely.

Set of integers = {... -3, -2, -1, 0, 1, 2, 3 ..}

- The set of integers includes both positive and negative whole number along with 0.

Prime numbers = [2, 3, 5, 7]

A whole number greater than 1 that can't be exactly divide by any whole num other than itself and 1.

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Tuple

- A tuple is a collection of elements represented within the symbol pair [and]. A tuple can contain duplicate elements.

- Note; a tuple is different to open intervals. For example, the tuple (a, b) is different to the open interval (a, b)

eg:- A = [3, 6, 7, 6, 3]

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Cartesian product of set

- Cartesian product of sets are represented as $A \times B$, $A \times B \times C$

- A cartesian product of set is also a set th consists of all possible ordered tuples of elements such that each element of tuple is take from each set in that order.

- Definition.

- The cartesian products of sets $A_1, A_2, \ldots, A_n$ is defined as,

$A_1 \times A_2 \times \ldots \times A_n = \{(x, y, \ldots z) \mid x \in A_1, y \in A_2, A_n \ldots, z \in A_$

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Relations

- Relations between sets, describe the connection or associate between different sets.
- It represent like a subset of set of tuples.

eg:- $\{(x,y): x \in \mathbb{N} \text{ and } y \in \mathbb{N} \mid x < 4 \text{ and } y < x\}$

$R = \{(3,2), (2,1), (3,1)\}$

- When a relation is a set of pairs (2 elements) it is a binary relation.
- When a relation is a set of triples (3 elements) it is a tenary relation.

eg: → Binary relation

$A = \{1,2,3,4,5,6,7,8\}$

$R = \{(1,2), (2,4), (3,6), (4,8)\}$

$\{(x,y): x \in \mathbb{N} \text{ and } y \in \mathbb{N} \mid x \leq 4 \text{ and } y = 2x\}$

Relation.

This relation should be same for all the elements.

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eg:- Tenary Relation.

$B = \{1, 2, 3, 4, 5, 6, 7\}$

$R = \{(1, 2, 3), (2, 4, 6), (2, 3, 5), (3, 4, 7)\}$

$\{(x, y, z) : x \text{ and } y \text{ and } z \in \mathbb{N} \mid z = x + y \text{ and } x < 4$

relation

Symmetry

- R is a relation and (a, b) is a pair.

- Symmetry means →

  If (a, b) is related by R, then (b, a) is also related by R

  $(a, b) \in R$ implies $(b, a) \in R$

eg:- Equal relation is a symmetry relation.

If $a = b$ then $b = a$

- This is for binary relations.

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Transtivity

- R is a relation and (a,b) and (b,c) are pairs.

- Transtivity means →

  If (a,b) is related by R, and (b,c) is also related by R, then (a,c) must related by R

$(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$

eg:- Greater than

If $x < y$ and $y < z$

then $x < z$

This is for ternary relations.

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Reflexivity

- Every element in the set related to itself.
  $(a, a) \in R$

eg:- (01) Greater or equal ($\leq$)

Lets take x, y and z

- $x \leq x \rightarrow$ This is correct ($x = x$)
  So we can say it is reflexive.

- Also $x \leq y$ and $y \leq z$ then $x \leq z$
  So we can say this is transitive as well.

- But, $x \leq y \neq y \leq x$ that means "$\leq$" from this we can't connect both sides.
- So this is not symmetry.

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Equal (=)

- $x = x$ → This is true. So it is reflexive.
- $x = y$ and $y = z$ then $x = z$. Also transitive.
- $x = y$ imply $y = x$. So it's symmetrical.

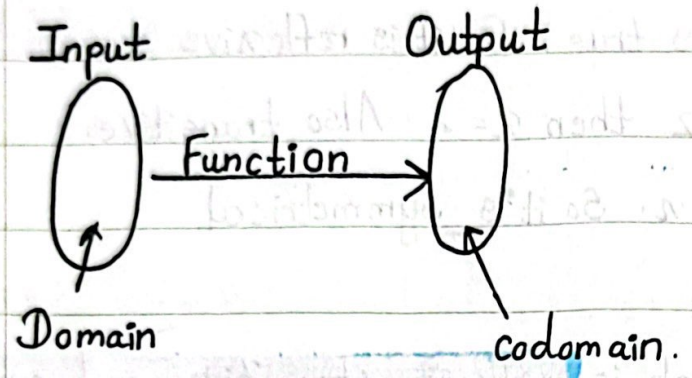
A relation which is reflexive, transitive and symmetrical is called Equivolence.

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Function



- A function is a rule that assigns each input to exactly one output.

- The set of inputs → Domain
- The set of outputs → Codomain (range of input)

eg:- a) $\{(0,-5),(1,-4),(2,-3),(3,-2),(4,-1)\}$

The domain → $\{0,1,2,3,4\}$

The codomain → $\{-5,-4,-3,-2,-1\}$

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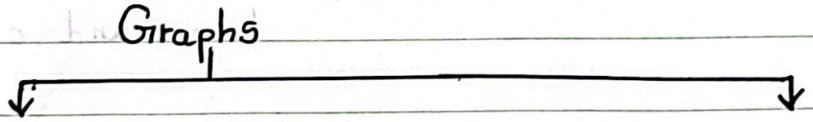
Graphs

- A graph is a finite set of nodes and edges between node

- Formally, a graph G is a structure (V, E) where,
  - V is a finite set of nodes
  - E that is a set of directed or undirected pairs of node x and y from V.

- The nodes represent the entities.
- The edges represent the relationships between entities x :

eg :- "$x < y$"
"x is larger than y"
"x is faster than y".



Directed (x, y)

- If the direction of the edge matter, we show the directions
- G is a directed graph, structure is (V, E)
  V → finite set of nodes
  E → A set of ordered pair

Undirected $\{x, y\}$

- If the direction of the ed doesn't matter, we don't sho
- G is a undirected graph. structure is (V, E)
  V → set of nodes
  E → set of unordered pair

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Directed Graph

$\{(x, y) : x \in V, y \in V\}$



Undirected Graph

$\{\{x, y\} : x \in V, y \in V\}$

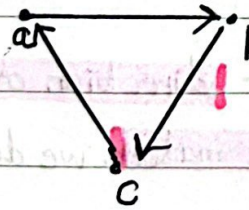


ex: i) $G = (V, E)$

$V = \{a, b, c\}$ → that has 3 nodes

$E = \{(a, b), (b, c), (c, a)\}$ → They connect a to b, b to c and c to a

This is a directed graph.



- curly brackets "{}" to represt edges means that graph is a undirected graph. While "()" means it is a directed graph.

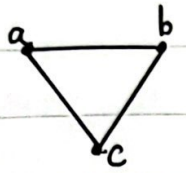
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(ii) $G = (V, E)$

$V = \{a, b, c\}$ → that has 3 nodes

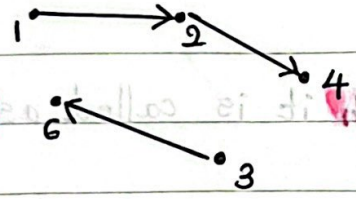
$E = \{\{a, b\}, \{b, c\}, \{c, a\}\}$ → They connected a and b, b and c, c and a



This is an undirected graph.

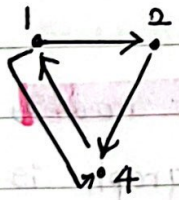
(iii) $V = \{1, 2, 3, 4, 5, 6\}$

$E = \{(1, 2), (2, 4), (5, 6)\}$



There is a pattern represented here.

(iv)



G_3

$V = \{1, 2, 3, 4\}$

$E = \{(1, 2), (2, 4), (4, 1), (1, 4), (3, 3)\}$.

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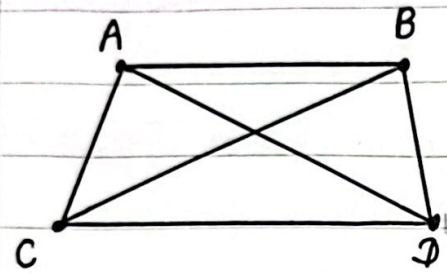
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Cycle in a graph

- A cycle in a graph G is a path where the last node is same as the first node.

- Consider the below graph

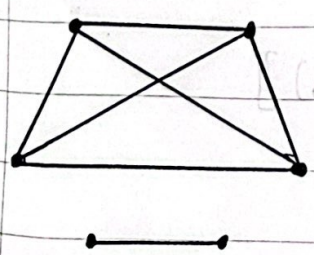


- A, B, C, A is a cycle
- A, D, C, A is a cycle.

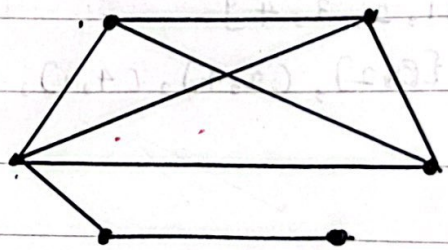
- If there is no cycle in the graph it is called as acyclic

Connectivity

- A graph is connected if for every pair of nodes, there is a path between them., Otherwise the graph is disconnected.



Disconnected



Connected.

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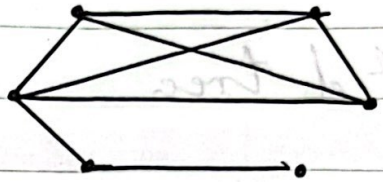
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Graph Cyclicity

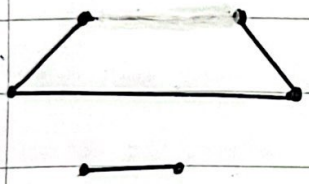
- A graph is cyclic if it has at least one cycle. Otherwise it is acyclic.



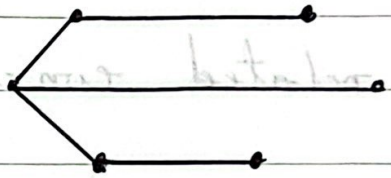
Disconnected and cyclic



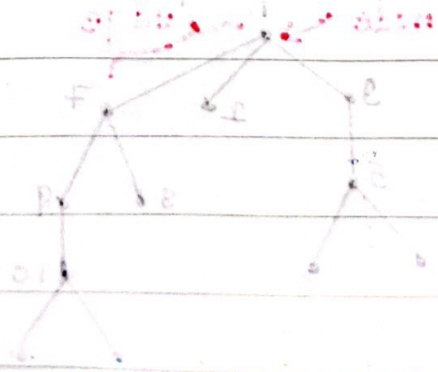
Connected and cyclic



Disconnected and acyclic



Connected and acyclic



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Trees

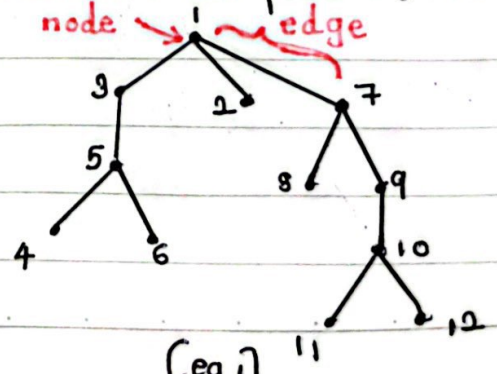
- A tree is a connected acyclic undirected graph.
- In a tree any two nodes are connected by exactly one path.

Rooted tree

- A rooted tree is a tree where one of the nodes is designated as the root node. (We cannot have two roots in a tree)

Tree related concepts

- The nodes adjacent to node x and below x are called the children of x.
  and x is called their parents.
- A node that has no children is a leaf.
- The descendents of a node are : its children. (their children all the way down)
- The ancestors of a node are : its parent, its grandpare all the way to the root.



(eg:)

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Leaves

- A node that has no children is called a leaf.

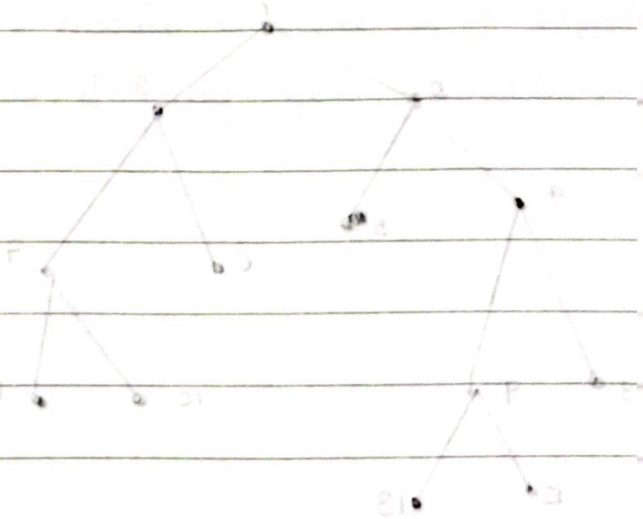
eg 1:- 2, 4, 6, 8, 11, 12

Concepts

- The depth of a node is the number of edges from the root to that node.
- The depth (or height) of a rooted tree is the depth of the lowest leaf.

From eg 1:-

- Depth of node 10 → 3
- Depth of the tree → 4



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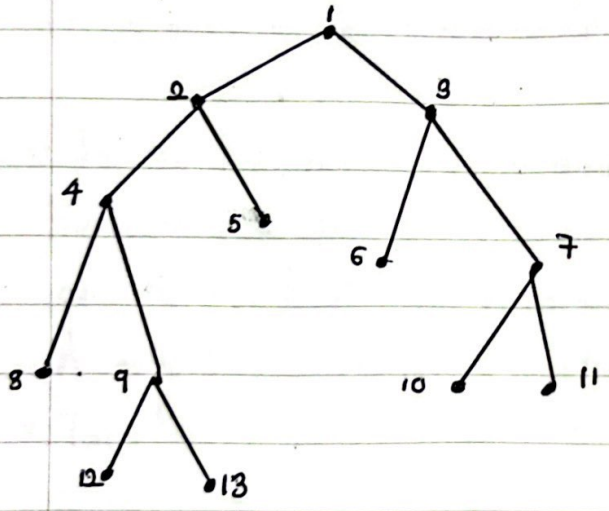
Date No

Binary trees

- A tree is a binary tree if every node has at most two children (one is also possible, but if it has more than two children it's a non binary tree)

- The children of any node in a binary tree are ordered into a left child and a right child.

- A node can have a left and a right child, a left child only, a right child only or no children.



(This is a binary tree).

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